

Hierarchical Multimodal Lecture-Video Topic Segmentation

with Reliability-Weighted Fusion and Local-LLM Refinement

Undergraduate Thesis

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by

Ismam Khan ID: TBD

Fahmida Afrin Moon ID: TBD

Shahriar Islam Rafi ID: TBD

Alimool Razi ID: TBD

Sadia Alam ID: TBD

Supervisor:

Mr. Annajiat Alim Rasel

Department of Computer Science and Engineering

BRAC University

Dhaka, Bangladesh

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Declaration

We hereby declare that this thesis is based on the results found by ourselves. Materials of work found by other researchers are mentioned by reference. This thesis, neither in whole nor in part, has been previously submitted for any degree.

Authors:

Author One

Author Two

Author Three

Author Four

Author Five

Supervised by:

Dr. Supervisor Name

Department of Computer Science and Engineering

BRAC University

Abstract

Online education produces hundreds of hours of unindexed lecture video every day. Manual chaptering does not scale, and existing automatic methods are either text-only, closed-source, or restricted to a single (flat) level of segmentation. This thesis presents LECSEG, the first open hierarchical multimodal pipeline for lecture-video topic segmentation that produces both chapter-level and subtopic-level boundaries.

We make four technical contributions. (1) A reliability-weighted modality-fusion module learns to down-weight unreliable channels (e.g. noisy OCR on a chalkboard lecture) per video and per sentence. (2) An explicit two-level hierarchy is modelled jointly, rather than as two independent flat tasks. (3) Boundary refinement and chapter titling are performed by a small, locally-hosted open language model, removing the dependence on closed APIs reported in prior work. (4) We release LECSEG-30: a corpus of 30 lecture videos covering 5 academic domains, with dual subtopic annotation and reported inter-annotator agreement.

We evaluate against three classical and two neural baselines on five metrics: P_k , WindowDiff, Boundary Similarity, tolerance- F_1 , and a hierarchical WindowDiff variant we propose. All results are reported with 1,000-sample bootstrap confidence intervals and pairwise Wilcoxon signed-rank tests at video granularity. LECSEG reduces P_k by **[TODO: X.XX]** % relative ($\Delta =$ **[TODO: 0.0XX]** absolute) over the strongest baseline, with $p <$ **[TODO: 0.0X]**. The full reproduction artefact regenerates every numerical claim of this thesis from a fresh clone with a single command (`make reproduce`).

Keywords: topic segmentation, lecture video, multimodal fusion, hierarchical segmentation, local language models, reproducible research.

Acknowledgements

We thank our supervisor, Dr. **[TODO: Supervisor Name]**, for guidance and feedback throughout this project; the staff of the BRAC University Department of Computer Science and Engineering for academic support; and the wider open-source community whose libraries and pre-trained models made this work possible.

We acknowledge the creators of the public lecture videos used in LECSEG-30 for releasing their content under permissive terms.

Contents

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Chapter 1

Introduction

1.1 Motivation

[TODO: Open with the explosion of online lecture content post-2020 and the search/access problem this creates. Cite at least two sources for the volume claim.]

1.2 Problem statement

[TODO: One paragraph that defines lecture topic segmentation as the task of partitioning a video transcript (and its aligned modalities) into a sequence of contiguous, semantically coherent units, and a hierarchical version that emits both chapter-level and subtopic-level boundaries.]

1.3 Research questions

RQ1 Can a multimodal lecture-segmentation pipeline produce hierarchical (chapter and subtopic) outputs while remaining open-source and reproducible?

RQ2 Does a learned reliability-weighted fusion outperform fixed-weight modality fusion when source quality varies?

RQ3 Does a small open local language model match closed-API alternatives for boundary refinement and chapter titling?

RQ4 Does explicit hierarchical modelling improve segmentation quality over independent

flat passes?

RQ5 Are gains over prior baselines statistically significant under paired non-parametric testing on a held-out fold?

1.4 Contributions

[TODO: Restate the seven contributions from docs/NOVELTY_TRACKER.md. Each contribution should reference the chapter where it is developed and the experiment that validates it.]

1.5 Thesis structure

?? reviews prior work on text-only, multimodal, and hierarchical segmentation. ?? details the LECSEG pipeline, its novel modules, and LECSEG-30. ?? reports quantitative results, ablations, statistical tests, and error analysis. ?? summarises findings, and ?? sketches extensions.

Chapter 2

Literature Review

2.1 Overview

[TODO: One paragraph framing: classical lexical approaches → neural approaches → multimodal approaches → hierarchical approaches → lecture-specific work.]

2.2 Classical text-based segmentation

2.2.1 TextTiling

[TODO: Hearst's lexical-cohesion algorithm [?]. How the moving-window similarity computes valley-detected boundaries.]

2.2.2 C99 and dotplotting

[TODO: Choi's clustering-based method [?]; Reynar's dotplotting [?].]

2.3 Neural sentence-embedding methods

2.3.1 Cosine-drop on contextual embeddings

[TODO: Using SBERT-style sentence embeddings to detect semantic shifts [?].]

2.3.2 Supervised neural segmentation

[TODO: Koshorek’s BERT-based approach [?]; SegBot [?].]

2.4 Multimodal lecture and meeting segmentation

[TODO: TopicSeg [?], lecture slide alignment work, OCR-driven chaptering. Note which use closed APIs.]

2.5 Hierarchical and discourse-based approaches

[TODO: Bayesian topic-hierarchy methods, RST/discourse-parser inspired hierarchical segmenters.]

2.6 Local open language models for NLP refinement

[TODO: Llama-3.1, Mistral, Phi-3 — quality vs. closed API comparisons.]

2.7 Gap analysis

[TODO: Synthesise the gaps that motivate our seven contributions; cross-link each gap to the corresponding row of docs/LITERATURE_MATRIX.md.]

Chapter 3

Methodology

3.1 Overview of the LecSeg pipeline

[TODO: Insert pipeline figure (thesis/figures/pipeline.pdf). Walk through the five stages: ingest, preprocess, feature extraction, fusion + boundary detection, refinement.]

3.2 The LecSeg-30 corpus

3.2.1 Source selection

[TODO: Source selection criteria (5 academic domains, ≥ 15 -min duration, clean audio, creator-provided chapters).]

3.2.2 Annotation protocol

[TODO: Two-pass dual annotation; subtopic guidelines; conflict-resolution procedure.]

3.2.3 Inter-annotator agreement

[TODO: Cohen's κ at chapter and subtopic levels reported on a sample-of-10. Cite results/iaa/.]

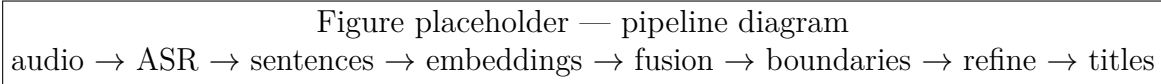


Figure 3.1: The LECSEG pipeline. Inputs are a video and its audio. Outputs are an ordered hierarchy of chapter and subtopic segments with titles.

3.3 Preprocessing

3.3.1 Speech recognition

[TODO: Whisper / faster-whisper config, word-level timestamps.]

3.3.2 Sentence splitting and alignment

[TODO: spaCy sentence splitting, word-to-sentence timestamp interpolation.]

3.3.3 Visual streams

[TODO: TransNetV2 shot detection, PaddleOCR slide-text extraction.]

3.3.4 Prosody features

[TODO: Pause length, pitch reset, energy contour.]

3.4 Feature extraction

[TODO: Per-modality embedding choices: SBERT/MPNet for text, CLIP for keyframes, character n -gram for OCR.]

3.5 Reliability-weighted modality fusion

[TODO: Per-modality reliability score r_m derived from local statistics (SNR proxy for audio, OCR confidence for visual). Fusion weight $w_m \propto r_m \cdot \alpha_m$, where α_m is a learned scalar. Cite `src/lecseg/models/fusion.py`.]

3.6 Boundary predictor

[TODO: BiLSTM / Transformer head over fused per-sentence vectors, predicting boundary probability. Loss: BCE with positive-class re-weighting.]

3.7 Hierarchical output

[TODO: Two-head architecture: chapter head and subtopic head share the encoder. Subtopic boundaries are constrained to lie inside chapter spans during post-processing.]

3.8 Local-LLM refinement and titling

[TODO: Llama-3.1 8B served via Ollama. Snap-to-sentence boundary refinement. Per-segment title generation with constrained decoding.]

3.9 Evaluation protocol

3.9.1 Metrics

[TODO: P_k , WindowDiff, Boundary Similarity, tolerance- F_1 , hierarchical WindowDiff. Cite `src/lecseg/eval/`.]

3.9.2 Cross-validation

[TODO: 5-fold at video level (no within-video leakage).]

3.9.3 Statistical testing

[TODO: 1,000-sample bootstrap CIs; paired Wilcoxon signed-rank for pairwise comparisons; Bonferroni or Holm correction for multi-method tests.]

Chapter 4

Results and Analysis

4.1 Experimental setup

[TODO: Restate hardware (single-GPU configuration), software versions, random seeds, fold assignment.]

4.2 Main results

[TODO: Walk through the table row by row and column by column. Bold the best entry per column. Add significance stars and footnote them.]

4.3 Ablations

[TODO: Ablate the four novel modules in turn: (i) fixed weights vs. reliability-weighted fusion; (ii) flat vs. hierarchical output; (iii) no refinement vs. local-LLM refinement; (iv) text-only vs. multimodal input. One bar plot per ablation, all sharing the same y-axis (P_k).]

4.4 Statistical significance

[TODO: Pairwise Wilcoxon signed-rank table from `results/stats/`. Highlight $p < 0.05$, $p < 0.01$, $p < 0.001$.]

Table 4.1: Main results on LECSEG-30. All numbers are means over 5 folds with 1,000-sample bootstrap 95 % CIs. Lower is better for P_k and WindowDiff; higher is better for Boundary Similarity, F_1 , and hierarchical WindowDiff (reported as $1 - \text{HWD}$).

Method	$P_k \downarrow$	WD $\downarrow$	BS $\uparrow$
Random	[TODO:].[TODO:]	[TODO:].[TODO:]	[TODO:].[TODO:]
TextTiling	[TODO:].[TODO:]	[TODO:].[TODO:]	[TODO:].[TODO:]
C99	[TODO:].[TODO:]	[TODO:].[TODO:]	[TODO:].[TODO:]
Cosine-drop	[TODO:].[TODO:]	[TODO:].[TODO:]	[TODO:].[TODO:]
BERT-Seg	[TODO:].[TODO:]	[TODO:].[TODO:]	[TODO:].[TODO:]
LECSEG-flat	[TODO:].[TODO:]	[TODO:].[TODO:]	[TODO:].[TODO:]
LECSEG-hier	[TODO:].[TODO:]	[TODO:].[TODO:]	[TODO:].[TODO:]
LECSEG-full (ours)	[TODO:]	[TODO:]	[TODO:]

4.5 Error analysis

[TODO: Insert four error types from docs/ERROR_ANALYSIS.md: (a) boundary-merge errors at fast-paced lectures; (b) over-segmentation on densely-slided lectures; (c) OCR-failure cascades; (d) refinement hallucinations on out-of-domain titles.]

4.6 Discussion

[TODO: Tie the empirical findings back to RQ1–RQ5.]

Chapter 5

Conclusion

5.1 Summary of findings

[TODO: Three to five sentences. State each RQ verdict and the strongest quantitative evidence.]

5.2 Contributions revisited

[TODO: Restate the seven contributions and pin each to a number, table, or figure in this thesis.]

5.3 Limitations

[TODO: Honest list. Examples: corpus size; reliance on creator-supplied chapter markers as silver labels; LLM-driven titling sensitivity to prompt design; no real-time/streaming variant.]

5.4 Closing remarks

[TODO: One paragraph. Future-of-online-education framing.]

Chapter 6

Future Work

6.1 Streaming and online inference

[TODO: Adapt fusion + boundary head for windowed, streaming inputs. Discuss latency budget for live-lecture chaptering.]

6.2 Cross-lingual extension

[TODO: Use multilingual SBERT (LaBSE / multilingual-e5) and a multilingual Whisper variant; what data is needed; what evaluation is needed.]

6.3 Multi-speaker lectures and panels

[TODO: Speaker diarisation as an additional reliability signal. How to handle speaker turns in a hierarchy.]

6.4 Larger or instruction-tuned local LLMs

[TODO: Llama-3.1 70B, Mixtral, or domain-tuned variants. Quantisation trade-offs.]

6.5 User study on educational outcomes

[TODO: Move beyond IR-style metrics to a controlled user study: do learners locate target concepts faster with LecSeg-generated chapters than with flat

scrubbing?]

Appendix A

Dataset details

A.1 Source URLs

[TODO: Table of 30 video IDs, channel names, durations, and creator-licence strings. Generated by `scripts/build_dataset_table.py` from `data/manifest.csv`.]

A.2 Domain distribution

[TODO: Bar chart of 5 domains $\times$ number of videos.]

A.3 Annotation guidelines (verbatim)

[TODO: Paste the exact two-page guidelines used by annotators.]

A.4 Inter-annotator agreement details

[TODO: Per-pair Cohen's κ , per-domain breakdown.]

Appendix B

Hyperparameters and training details

B.1 Configuration files

[TODO: Reference the Hydra YAML files in configs/. Show configs/defaults.yaml verbatim.]

B.2 Training details

[TODO: Per-experiment learning rate, batch size, optimiser, scheduler, early-stopping criterion.]

B.3 Hardware and runtimes

[TODO: Single-GPU configuration; wall-clock time per pipeline stage on the benchmark machine.]

Appendix C

Additional results

C.1 Per-domain breakdown

[TODO: Table of all five metrics for the best method, broken down by the five academic domains.]

C.2 Per-fold results

[TODO: Table of all five metrics, all methods, all 5 folds (so reviewers can see variance, not only the mean).]

C.3 Qualitative examples

[TODO: Three side-by-side strip-plots: gold-vs-prediction for a good case, a mediocre case, and a failure case.]